

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY



Department of Computer Science and Engineering

Course No: CSE 2104

Course Title: Data Structures Lab

Assignment No: 02

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Submitted by,

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Lab Section: B1

Academic Semester: Spring 2025

Program: BSc in CSE

Question:

You are given a list of books, each with a title, author and publication year. Your task is to sort

the books by the publication year in descending order (most recent first) and if two books have

the same publication year, sort them lexicographically by author in ascending order. If both the

publication year and author are the same, sort the books lexicographically by title in ascending

order. Implement this sorting using the Selection Sort algorithm. The input consists of an integer

N (the number of books), followed by N lines of book details, where each line contains the title,

author and publication year. Output the books in the sorted order, each on a new line in the

format: "Title, Author, Publication Year".

Input Format

- The first line contains an integer N, the number of books ($1 \leq N \leq 1000$).
- Each of the next N lines contains a book's information in the format:
 - Title, Author, Publication Year
- The Title is a string ($1 \leq \text{length of title} \leq 100$)
- The Author is a string ($1 \leq \text{length of author name} \leq 100$)
- The Publication Year is an integer ($1900 \leq \text{year} \leq 2025$)

Output Format

Print the books in sorted order based on the criteria above, one book per line in the following

format:

- Title, Author, Publication Year

Sample Input Sample :

5

Sapiens, Yuval Noah Harari, 2011

21 Lessons for the 21st Century, Yuval Noah
Harari, 2018

Educated, Tara Westover, 2018

Becoming, Michelle Obama, 2018

Money: Vintage Minis, Yuval Noah Harari,
2018

Output:

Becoming, Michelle Obama, 2018

Educated, Tara Westover, 2018

21 Lessons for the 21st Century, Yuval Noah
Harari, 2018

Money: Vintage Minis, Yuval Noah Harari,
2018

Sapiens, Yuval Noah Harari, 2011

Code:

```
#include<bits/stdc++.h>
```

```
using namespace std;
```

```
bool isBest(int a,int  
b,vector<int>&year,vector<string>&author,vector<string>&title)
```

```
{  
    if(year[a]!=year[b])  
    {  
        return year[a]>year[b];  
    }  
    else if(author[a]!=author[b])  
    {  
        return author[a]<author[b];  
    }  
    else  
    {  
        return title[a]<title[b];  
    }  
}
```

```
int main()  
{  
    int N;  
    cin>>N;  
    cin.ignore();  
  
    vector<string>title(N),author(N);
```

```
vector<int>year(N);
```

```
for(int i=0; i<N; i++)
```

```
{
```

```
    string pubYear;
```

```
    getline(cin,title[i],',');
```

```
    getline(cin,author[i],',');
```

```
    getline(cin,pubYear);
```

```
    year[i]=stoi(pubYear);
```

```
}
```

```
for(int i=0; i<N; i++)
```

```
{
```

```
    int best=i;
```

```
    for(int j=i+1; j<N; j++)
```

```
    {
```

```
        if(isBest(j,best,year,author,title))
```

```
        {
```

```
            best = j;
```

```
        }
```

```
    }
```

```
    swap(year[i],year[best]);
```

```

        swap(author[i],author[best]);
        swap(title[i],title[best]);
    }

    for(int i=0; i<N; i++)
    {
        cout << title[i] << "," << author[i] << "," << year[i] << endl;
    }

    return 0;
}

```

Sample Input Sample :

5

Sapiens,Yuval Noah Harari,2011

21 Lessons for the 21st Century,Yuval Noah Harari, 2018

Educated,Tara Westover,2018

Becoming,Michelle Obama,2018

Money: Vintage Minis,Yuval Noah Harari,2018

Output:

Becoming,Michelle Obama,2018

Educated,Tara Westover,2018

21 Lessons for the 21st Century,Yuval Noah Harari,2018

Money: Vintage Minis, Yuval Noah Harari, 2018

Sapiens, Yuval Noah Harari, 2011